

High Performance Stereo Audio DAC

FEATURES

- High performance and low power multi-bit delta-sigma audio DAC
- 110 dB signal to noise ratio, -80 dB THD+N
- 24-bit, 8 to 96 kHz sampling frequency
- Integrated headphone driver with capless option
- Differential output for higher SNR and CMRR
- I²S/PCM master or slave serial data port
- 256/384Fs, USB 12/24 MHz and other non standard audio system clocks
- I²C interface
- 7-band fully adjustable EQ
- Dynamic range compression
- Playback signal feedback
- Pop and click noise suppression
- 1.8V to 3.3V operation

APPLICATIONS

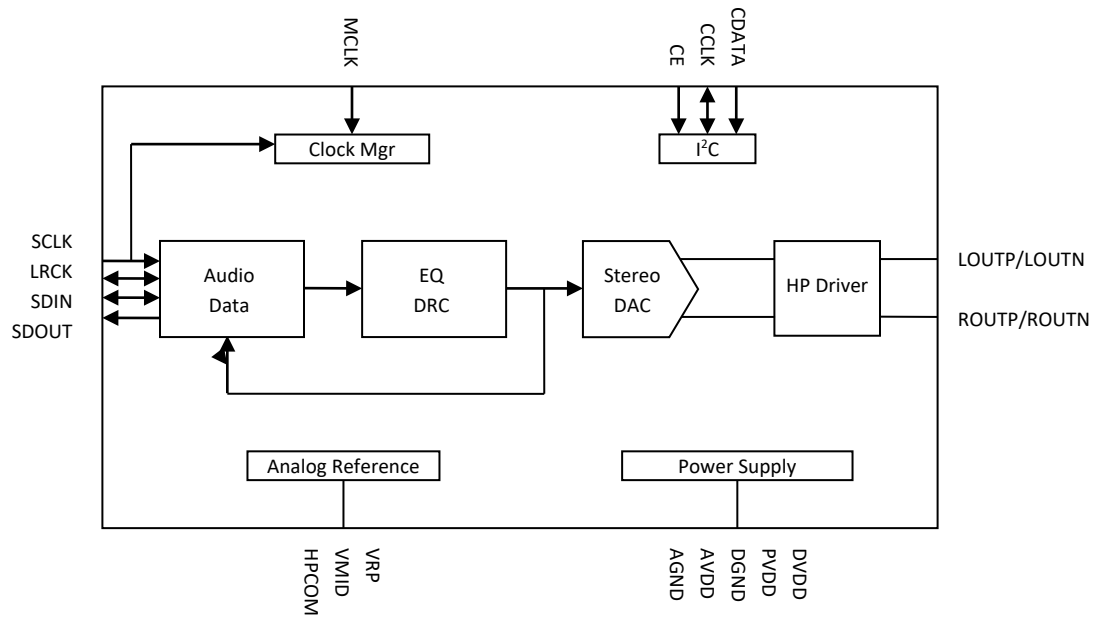
- Headphone
- Speaker
- TV
- Portable audio devices

ORDERING INFORMATION

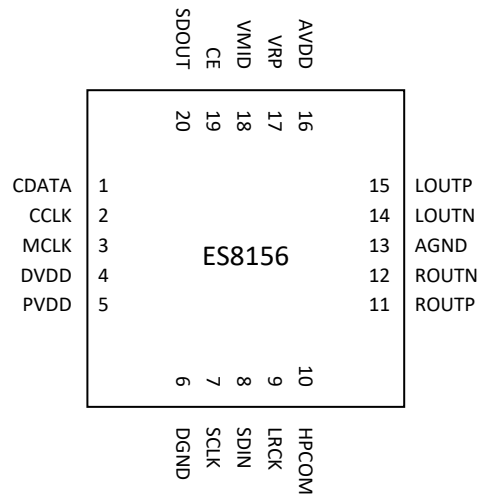
ES8156 -40°C ~ +125°C
QFN-20

1.	BLOCK DIAGRAM.....	3
2.	PIN OUT AND DESCRIPTION.....	4
3.	TYPICAL APPLICATION CIRCUIT.....	5
4.	HARDWARE MODE.....	7
5.	CLOCK MODES AND SAMPLING FREQUENCIES.....	7
6.	MICRO-CONTROLLER CONFIGURATION INTERFACE	7
7.	DIGITAL AUDIO INTERFACE.....	9
8.	ELECTRICAL CHARACTERISTICS	10
	ABSOLUTE MAXIMUM RATINGS.....	10
	RECOMMENDED OPERATING CONDITIONS	10
	DAC ANALOG AND FILTER CHARACTERISTICS AND SPECIFICATIONS.....	10
	DC CHARACTERISTICS	11
	I ² C SWITCHING SPECIFICATIONS (SLOW SPEED MODE/HIGH SPEED MODE)	11
	SERIAL AUDIO PORT SWITCHING SPECIFICATIONS	12
9.	PACKAGE (UNIT: MM).....	13
10.	CORPORATE INFORMATION	15
11.	IMPORTANT NOTICE AND DISCLAIMER.....	15

1. BLOCK DIAGRAM

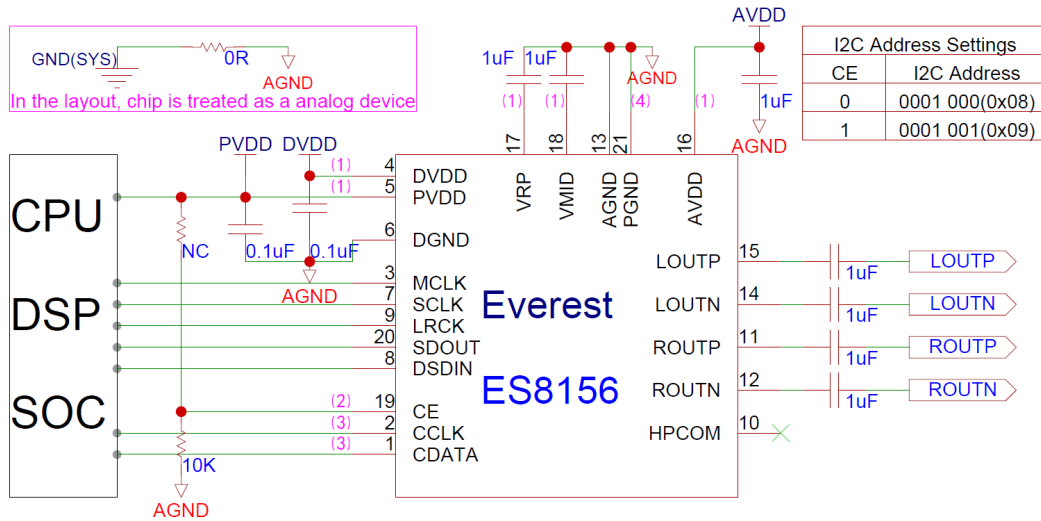


2. PIN OUT AND DESCRIPTION

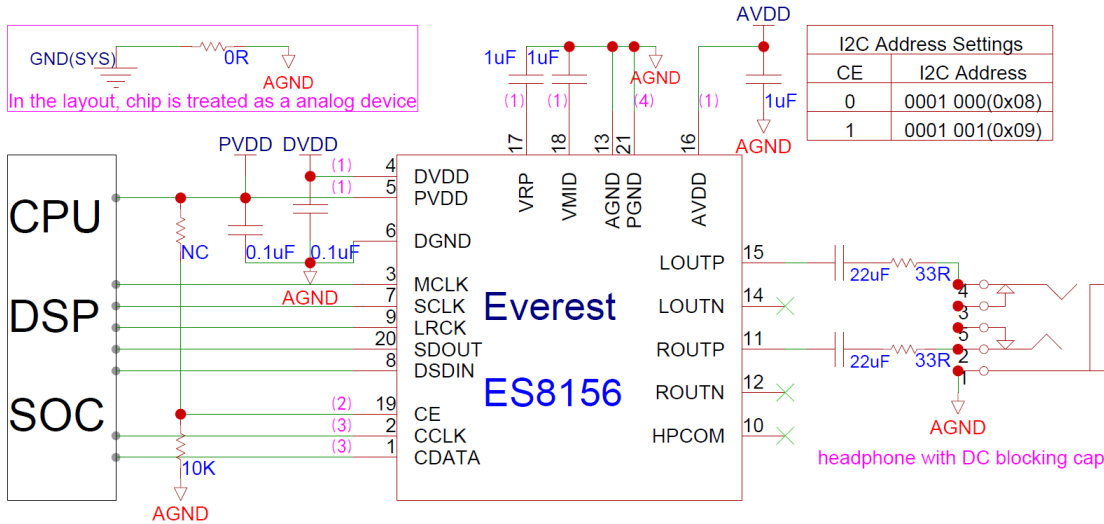


Pin Name	Pin number	Input or Output	Pin Description
CDATA, CCLK, CE	1, 2, 19	I/O, I, I	I ² C clock, data, address
MCLK	3	I	Master clock
SCLK	7	I/O	Serial data bit clock/DMIC bit clock
SDIN	8	I	DAC serial data input
LRCK	9	I/O	Serial data left and right channel frame clock
SDOUT	20	O	Playback signal feedback
LOUTP, LOUTN	15, 14	O	Left channel differential analog output
ROUTP, ROUTN	11, 12	O	Right channel differential analog output
HPCOM	10	Analog	Virtual ground for capless headphone (Only available in software mode)
PVDD	5	Analog	Power supply for the digital input and output
DVDD, DGND	4, 6	Analog	Digital power supply
AVDD, AGND	16, 13	Analog	Analog power supply
VMID	18	Analog	Filtering capacitor connection
VRP	17	Analog	Filtering capacitor connection

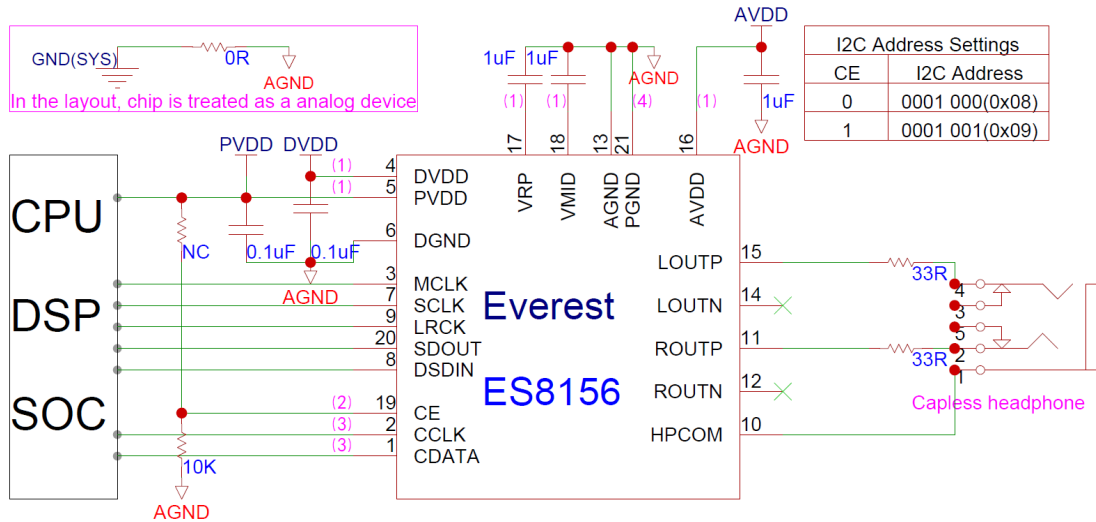
3. TYPICAL APPLICATION CIRCUIT



- (1) For optimal performance, the filter capacitor should be placed on the same layer and as close as possible to the pin it is filtering
 (2) Each address pin(CE) must be tied to either PVDD(logic high) or AGND (logic low) using an external resistor
 (3) Reservation(预留) of RC filter circuit in the I2C signal path are recommended in case I2C signal integrity are not ideal(理想)
 (4) PIN21:PGND(on the underside of the chip)must be electrically tied to the ground



- (1) For optimal performance, the filter capacitor should be placed on the same layer and as close as possible to the pin it is filtering
 (2) Each address pin(CE) must be tied to either PVDD(logic high) or AGND (logic low) using an external resistor
 (3) Reservation(预留) of RC filter circuit in the I2C signal path are recommended in case I2C signal integrity are not ideal(理想)
 (4) PIN21:PGND(on the underside of the chip)must be electrically tied to the ground



4. HARDWARE MODE

The device works either in hardware mode (HW mode) or software mode (I²C mode). The default is hardware mode. Software mode is enabled by setting bit 2 of configuration register 0x02.

In HW mode, LRCK and SCLK are supplied externally. LRCK and SCLK must be synchronously derived from the system clock with some specific rates. The device can auto detect MCLK/LRCK ratio according to Table 1. The device only supports the MCLK/LRCK ratios listed in Table 1. The SCLK/LRCK ratio is normally 64.

Table 1 Slave Mode Sampling Frequencies and MCLK/LRCK Ratio

Speed Mode	Sampling Frequency	MCLK/LRCK Ratio
Single Speed	8kHz – 50kHz	32, 64, 96, 128, 192, 256, 384, 512, 640, 768, 1024, 1152, 1280, 1536

5. CLOCK MODES AND SAMPLING FREQUENCIES

In software mode, the device supports standard audio clocks (32Fs, 64Fs, 128Fs, 256Fs, 384Fs, 512Fs, etc), USB clocks (12/24 MHz), and some common non standard audio clocks (16 MHz, 25 MHz, 26 MHz, etc).

According to the serial audio data sampling frequency (Fs), the device can work in two speed modes: single speed mode or double speed mode. In single speed mode, Fs normally ranges from 8 kHz to 48 kHz, and in double speed mode, Fs normally range from 64 kHz to 96 kHz.

The device can work either in master clock mode or slave clock mode. In slave mode, LRCK and SCLK are supplied externally, and LRCK and SCLK must be synchronously derived from the system clock with specific rates. In master mode, LRCK and SCLK are derived internally from device master clock.

6. MICRO-CONTROLLER CONFIGURATION INTERFACE

The device supports standard I²C micro-controller configuration interface. External micro-controller can completely configure the device through writing to internal configuration registers.

I²C interface is a bi-directional serial bus that uses a serial data line (CDATA) and a serial clock line (CCLK) for data transfer. The timing diagram for data transfer of this interface is given in Figure 1a and Figure 1b. Data are transmitted synchronously to CCLK clock on the CDATA line on a byte-by-byte basis. Each bit in a byte is sampled during CCLK high with MSB bit being transmitted firstly. Each transferred byte is followed by an acknowledge bit from receiver to pull the CDATA low. The transfer rate of this interface can be up to 400 kbps.

A master controller initiates the transmission by sending a “start” signal, which is defined as a high-to-low transition at CDATA while CCLK is high. The first byte transferred is the slave address. It is a seven-bit chip address followed by a RW bit. The chip address must be 0001 00x, where x equals CE (input pin: 1 being connected to supply and 0 being connected to ground). The RW bit indicates the slave data transfer direction. Once an acknowledge bit is received, the data transfer starts to proceed on a byte-by-byte basis in the direction specified by the RW bit. The master can terminate the communication by generating a “stop” signal, which is defined as a low-to-high transition at CDATA while CCLK is high.

In I²C interface mode, the registers can be written and read. The formats of “write” and “read” instructions are shown in Table 2 and Table 3. Please note that, to read data from a register, you must set R/W bit to 0 to access the register address and then set R/W to 1 to read data from the register.

Table 2 Write Data to Register in I²C Interface Mode

	Chip Address	R/W		Register Address		Data to be written		
start	0001 00 CE	0	ACK	RAM	ACK	DATA	ACK	Stop

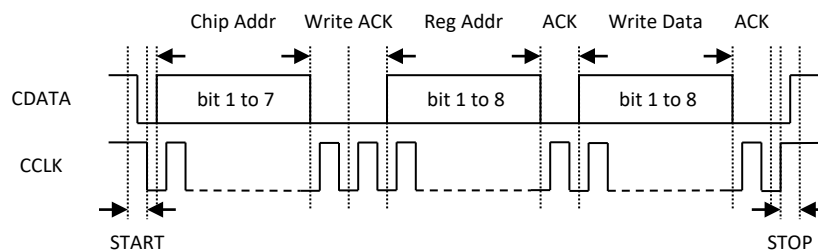


Figure 1a I²C Write Timing

Table 3 Read Data from Register in I²C Interface Mode

	Chip Address	R/W		Register Address		
Start	0001 00 CE	0	ACK	RAM	ACK	
	Chip Address	R/W		Data to be read		
Start	0001 00 CE	1	ACK	Data	NACK	Stop

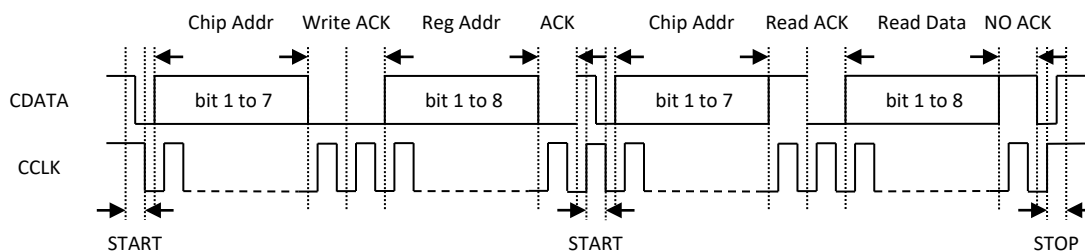


Figure 1b I²C Read Timing

7. DIGITAL AUDIO INTERFACE

The device provides many formats of serial audio data interface to the input or output through LRCK, SCLK and SDIN or SDOUT pins. These formats are I²S, left justified, right justified and DSP/PCM. SDIN is sampled by the device on the rising edge of SCLK. SDOUT is out on the falling edge of SCLK. The relationship of SDATA (SDIN or SDOUT), SCLK and LRCK with these formats are shown through Figure 2a to Figure 2d.

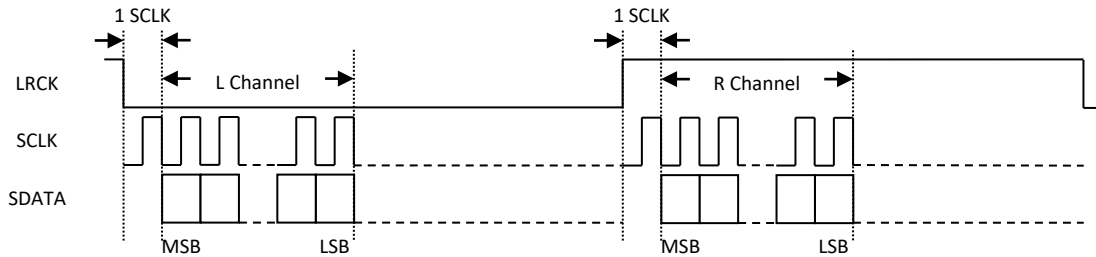


Figure 2a I²S Serial Audio Data Format

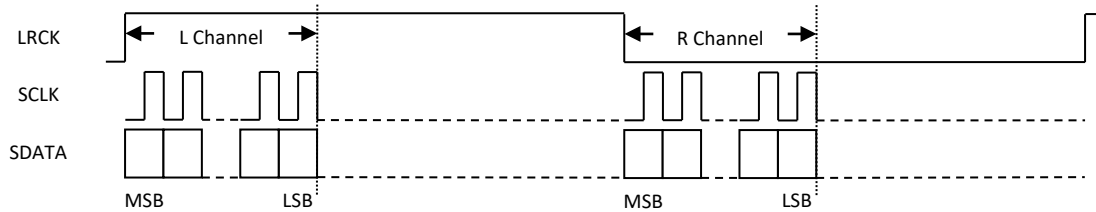


Figure 2b Left Justified Serial Audio Data Format

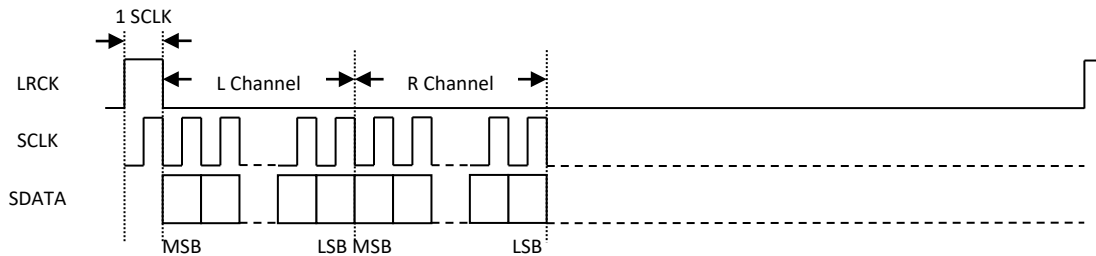


Figure 2c DSP/PCM Mode A Serial Audio Data Format

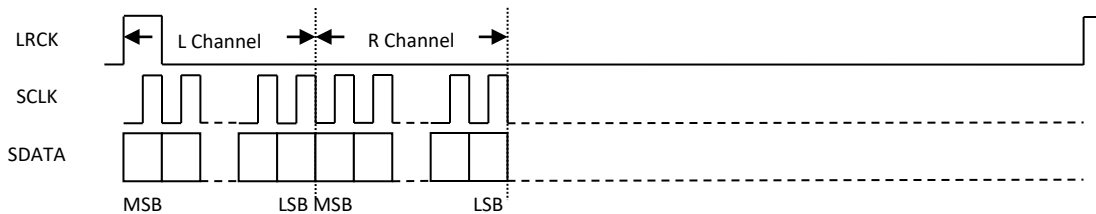


Figure 2d DSP/PCM Mode B Serial Audio Data Format

8. ELECTRICAL CHARACTERISTICS

ABSOLUTE MAXIMUM RATINGS

Continuous operation at or beyond these conditions may permanently damage the device.

PARAMETER	MIN	MAX
Analog Supply Voltage Level	-0.3V	+3.6V
Digital Supply Voltage Level	-0.3V	+3.6V
Analog Input Voltage Range	AGND-0.3V	AVDD+0.3V
Digital Input Voltage Range	DGND-0.3V	PVDD+0.3V
Operating Temperature Range	-40°C	+125°C
Storage Temperature	-65°C	+150°C

RECOMMENDED OPERATING CONDITIONS

PARAMETER	MIN	TYP	MAX	UNIT
DVDD (Note 1)	1.6	1.8/3.3	3.6	V
PVDD	1.6	1.8/3.3	3.6	V
AVDD (Note 2)	1.7	1.8/3.3	3.6	V

Note 1: The internal clock source can be MCLK or SCLK. When this internal clock source is multiplied by 4 or 8, its frequency must be greater than 1 MHz for 3.3V DVDD or 500 kHz for 1.8V DVDD.

Note 2: in hardware mode, AVDD must be higher than 3V.

DAC ANALOG AND FILTER CHARACTERISTICS AND SPECIFICATIONS

Test conditions are as the following unless otherwise specify: AVDD=3.3V, DVDD=3.3V, AGND=0V, DGND=0V, Ambient temperature=25°C, Fs=48 KHz, MCLK/LRCK=256.

PARAMETER	MIN	TYP	MAX	UNIT
DAC Performance				
Signal to Noise ratio (A-weight)	105	110	115	dB
THD+N	-85	-80	-78	dB
Channel Separation (1KHz)	105	110	115	dB
Interchannel Gain Mismatch		0.2	1	dB
Filter Frequency Response – Single Speed				
Passband	0		0.4535	Fs
Stopband	0.5465			Fs
Passband Ripple			±0.05	dB
Stopband Attenuation	53			dB
Filter Frequency Response – Double Speed				
Passband	0		0.4167	Fs
Stopband	0.7917			Fs
Passband Ripple			±0.005	dB
Stopband Attenuation	56			dB
Analog Output				

Full Scale Output (differential P and N)	$1.71 \cdot AVDD/3.3$	$2 \cdot 0.9 \cdot AVDD/3.3$	$1.89 \cdot AVDD/3.3$	Vrms
--	-----------------------	------------------------------	-----------------------	------

DC CHARACTERISTICS

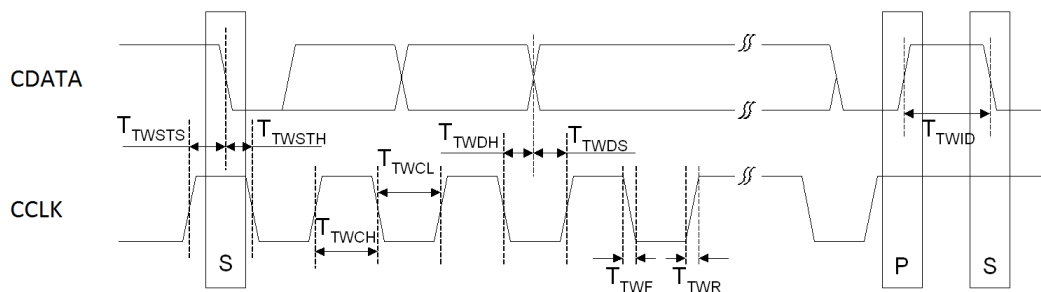
PARAMETER	MIN	TYP	MAX	UNIT
Normal Operation Mode				
DVDD=1.8V, PVDD=1.8V, AVDD=3.3V (Note 3a)		19		mW
DVDD=1.8V, PVDD=1.8V, AVDD=1.8V		6		
Power Down Mode (Note 3b)				
DVDD=1.8V, PVDD=1.8V, AVDD=3.3V		0		uA
Digital Voltage Level				
Input High-level Voltage	$0.7 \cdot PVDD$			V
Input Low-level Voltage			0.5	V
Output High-level Voltage		PVDD		V
Output Low-level Voltage		0		V

Note 3a: recommend VDDP and VDDD power supply turn on or off within 10 ms of each other; VDDD must be on when VDDA is on.

Note 3b: recommend all power supply on, entering low power through control register setting, then stopping input clock.

I²C SWITCHING SPECIFICATIONS (SLOW SPEED MODE/HIGH SPEED MODE)

PARAMETER	Symbol	MIN	MAX	UNIT
CCLK Clock Frequency	F_{CCLK}		100/400	KHz
Bus Free Time Between Transmissions	T_{TWID}	4.7/1.3		us
Start Condition Hold Time	T_{TWSTH}	4.0/0.6		us
Clock Low time	T_{TWCL}	4.7/1.3		us
Clock High Time	T_{TWCH}	4.0/0.6		us
Setup Time for Repeated Start Condition	T_{TWSTS}	4.7/0.6		us
CDATA Hold Time from CCLK Falling	T_{TWDH}		3.45/0.9	us
CDATA Setup time to CCLK Rising	T_{TWDS}	0.25/0.1		us
Rise Time of CCLK	T_{TWR}		1.0/0.3	us
Fall Time CCLK	T_{TWF}		1.0/0.3	us

Figure 3 I²C Timing

SERIAL AUDIO PORT SWITCHING SPECIFICATIONS

PARAMETER	Symbol	MIN	MAX	UNIT
MCLK frequency			49.2	MHz
MCLK duty cycle		40	60	%
LRCK frequency			100	KHz
LRCK duty cycle (Note 4)		40	60	%
SCLK frequency			26	MHz
SCLK pulse width low	T_{SLKL}	16		ns
SCLK Pulse width high	T_{SCLKH}	16		ns
SCLK falling to LRCK edge (master mode only)	T_{SLR}		10	ns
LRCK edge to SCLK rising (slave mode only)	T_{LSR}	10		ns
SCLK falling to SDOUT valid VDDD=3.3V VDDD=1.8V	T_{SDO}		16 39	ns
LRCK edge to SDOUT valid (Note 5) VDDD=3.3V VDDD=1.8V	T_{LDO}		11 25	ns
SDIN valid to SCLK rising setup time	T_{SDIS}	10		ns
SCLK rising to SDIN hold time	T_{SDIH}	10		ns

Note 4: one SCLK period of high time in DSP/PCM modes.

Note 5: only apply to MSB of Left Justified or DSP/PCM mode B.

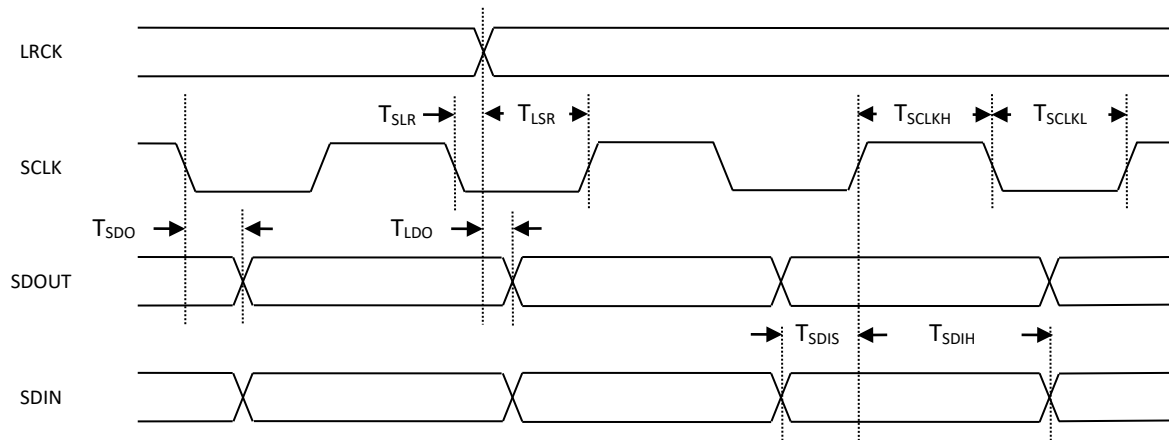
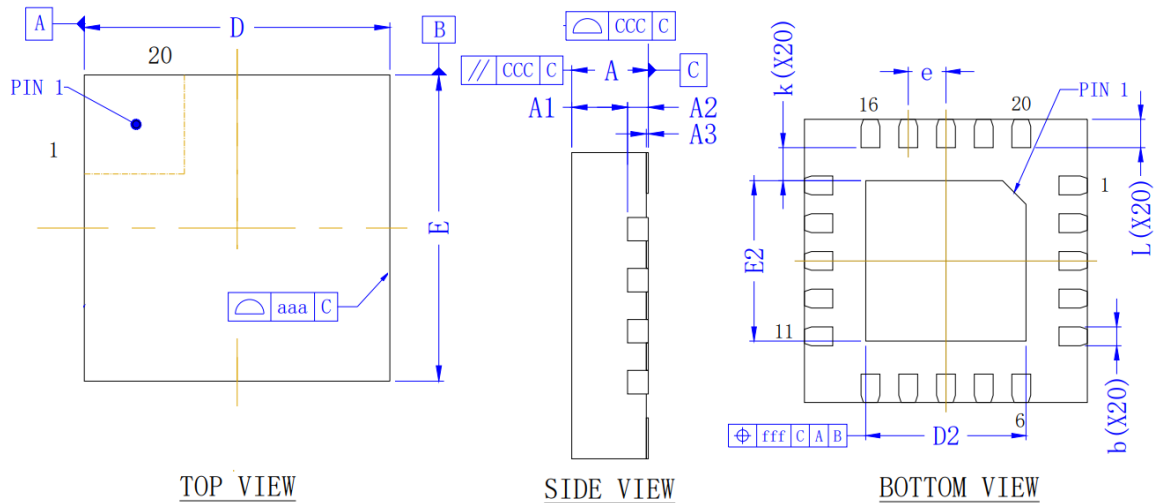


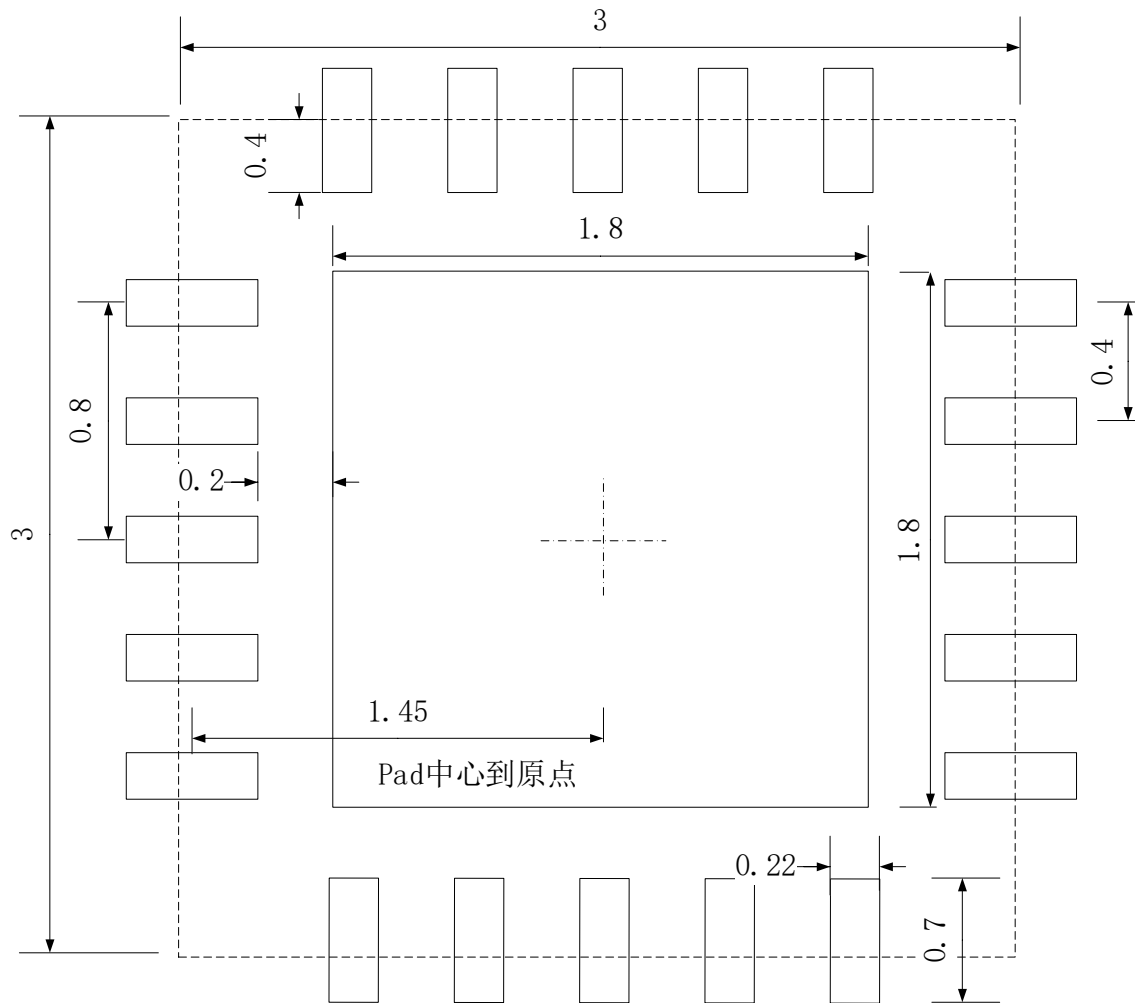
Figure 4 Serial Audio Port Timing

9. PACKAGE (UNIT: MM)

QFN-20L-3X3(P0.4T0.55) Package Outline Drawing



Item	Symbol	Minimum	Normal	Maximum
Total Thickness	A	0.50	0.55	0.60
Stand Off	A1	0.00	0.02	0.05
Molding Thickness	A2	---	0.40	---
LF Thickness	A3	0.152 REF		
Lead Width	b	0.15	0.20	0.25
Body Size	D	3 BSC		
	E	3 BSC		
Lead Pitch	e	0.4 BSC		
Exposed Pad Size	D2	1.60	1.70	1.80
	E2	1.60	1.70	1.80
Lead Length	L	0.20	0.30	0.40
Lead tip to Exposed Pad	K	0.35 REF		
Package Edge Tolerance	aaa	0.1		
Molding Flatness	ccc	0.1		
Coplanarity	eee	0.08		
Lead Offset	bbb	0.1		
Exposed Pad Offset	fff	0.1		



10. CORPORATE INFORMATION

Everest Semiconductor Co., Ltd.

No. 1355 Jinjihu Drive, Suzhou Industrial Park, Jiangsu, P.R. China, Zip Code 215021

苏州工业园区金鸡湖大道 1355 号国际科技园, 邮编 215021

Email: info@everest-semi.com



11. IMPORTANT NOTICE AND DISCLAIMER

Everest Semiconductor publishes reliable technical information about its products. Information contained herein is subject to change without notice. It may be used by a party at their own discretion and risk. Everest Semiconductor disclaims responsibility for any claims, damages, costs, losses, and liabilities arising out of your use of the information. This publication is not to be taken as a license to operate under any existing patents and intellectual properties.